

BUS713 Financial Analytics: Workshop 4

CAPM, Beta Estimation, WACC, and the Security Market Line

Builds on Module 3: beta and systematic risk, now used to price expected return

Module 3 introduced beta as a measure of systematic risk. Module 4 uses beta to answer a sharper question: given a stock's exposure to market risk, what return should it be expected to deliver? That question drives CAPM, the cost of equity in WACC, and the Security Market Line. Work through each problem yourself.

Task 1: What CAPM Says and Why It Matters

Scenario: *A colleague tells a client that a stock is a good buy because it "returned 15% last year," without mentioning risk at all. Your manager asks you to explain why that claim is incomplete.*

CAPM states that expected return equals the risk-free rate plus beta multiplied by the market risk premium, where the market risk premium is the market's expected return minus the risk-free rate. It prices only systematic risk, the kind that cannot be diversified away, because that is the only risk investors are compensated for holding.

1. In your own words, explain what each term in the CAPM formula represents (the risk-free rate, beta, and the market risk premium), and why CAPM builds expected return from exactly these two pieces.
2. Your colleague says "CAPM just tells us what a stock should return based on how risky it is." Is that accurate? Which kind of risk from Module 3, systematic or unsystematic, does CAPM actually price, and why does it ignore the other?
3. Using a risk-free rate of 4%, a market risk premium of 7%, and a beta of 1.3, calculate the CAPM-expected return for a stock. Explain what this number represents for an investor deciding whether the stock is worth buying.

Task 2: Estimating Beta with a Regression — Coca-Cola

Scenario: *You need Coca-Cola's beta for a valuation model, and your manager wants it estimated from data rather than pulled from a finance website.*

Beta is estimated as the slope of an OLS regression of a stock's excess return on the market's excess return. Excess return is the asset's return minus the risk-free rate over the same period.

4. Download monthly price data for KO and the S&P 500 (ticker "^GSPC") since 2015 using `tq_get()`, and compute monthly returns for both.
5. Using a monthly risk-free rate of 4% per year divided by 12, calculate excess returns for KO and for the market. Combine both into one data frame and run `lm(koke_excess ~ sp500_excess)` to estimate the regression.
6. Report KO's estimated beta, alpha, and R-squared from `summary()`. Is KO more or less volatile than the market? Based on the regression output, is alpha statistically distinguishable from zero?

Task 3: Comparing Beta Across Assets — Coca-Cola vs PepsiCo

Scenario: *A portfolio manager wants to know which of the two soft drink stocks carries more systematic risk before adjusting position sizes ahead of an expected downturn.*

7. Repeat the beta estimation process from Task 2 for PEP.
8. Compare the two betas. Which stock is more sensitive to market-wide movements? Does this match your intuition about consumer staples stocks generally carrying low betas?
9. If the market is expected to fall 10% in a recession scenario, use each stock's beta to estimate the expected decline in each stock's price. What does this imply for the portfolio manager deciding how much of each to hold going into a downturn?

Task 4: From Beta to CAPM Expected Return

Scenario: *You need cost-of-equity estimates for KO and PEP to feed into a WACC calculation next.*

10. Using the betas estimated in Tasks 2 and 3, a risk-free rate of 4%, and a market risk premium of 7%, calculate the CAPM-expected return for KO and for PEP.
11. Compare the two expected returns. Does the stock with the higher beta also have the higher required return? Is that a coincidence, or does it follow directly from the CAPM formula?

12. A client says they would only invest in KO if it can be expected to return at least 10% per year. Based on your CAPM estimate, would you recommend it? What would need to be true about KO's beta or the market risk premium for it to clear that bar?

Task 5: WACC — Combining Cost of Equity and Cost of Debt

Scenario: *You're valuing a potential acquisition target and need a single discount rate to use in the deal's net present value calculation.*

WACC weights the cost of equity and the after-tax cost of debt by each source's share of total firm value: $WACC = (E/V) \times \text{Cost of equity} + (D/V) \times \text{Cost of debt} \times (1 - \text{Tax rate})$, where E and D are market values and $V = E + D$.

13. A company has a market value of equity of \$600 million and market value of debt of \$400 million. Its cost of equity, from CAPM, is 11%, its pre-tax cost of debt is 6%, and its tax rate is 25%. Calculate the WACC.
14. Explain why the cost of debt is multiplied by $(1 - \text{tax rate})$ but the cost of equity is not.
15. A junior analyst uses the company's book value of debt and equity from the balance sheet instead of market values. Explain why this could produce a misleading WACC, and in which direction the error is likely to run for a company whose stock price has risen sharply since it issued its debt.

Task 6: The Security Market Line and CAPM's Limitations

Scenario: *A client asks whether KO looks like a good deal right now, and separately, whether CAPM can be trusted at all for the report you are preparing.*

The Security Market Line (SML) plots CAPM-expected return against beta. An asset plotting above the line is delivering more return than its beta justifies and is considered under-priced; an asset plotting below the line is overpriced.

16. Using the CAPM-expected return you calculated for KO in Task 4, suppose KO's actual realised average annual return over the last several years was higher than that CAPM prediction. Would KO plot above or below the SML? Is it under-priced or overpriced relative to the CAPM prediction?
17. A colleague asks, "if a stock plots above the SML, should we just buy it?" Explain what that position implies about the stock's current price, given the inverse relationship between price and return, and what key assumption about CAPM would need to hold for that conclusion to be reliable.
18. If CAPM were exactly correct, the alpha from the regression in Task 2 should be zero. In practice, estimated alphas vary and have been linked to factors like company size and value. In 3 to 4 sentences, explain what this implies about relying on CAPM alone for a client-facing risk-return report, and name one such factor.

Wrap-Up

You explained what CAPM prices and why, estimated beta for KO and PEP from real return data using OLS regression, converted beta into a CAPM-expected return, combined cost of equity and cost of debt into a WACC, and used the Security Market Line to reason about mispricing and CAPM's limitations. Together with Modules 1 to 3, you now have a full pipeline from raw prices to a discount rate.